SCMVC: Semantic Constraint-Based Spatial–Spectral Multiview Clustering for Hyperspectral Images

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***Abstract*—Cross-view consensus representation plays a crucial role in hyperspectral image (HSI) clustering. Recently, multiview contrastive clustering (MVCC) methods have leveraged contrastive loss to extract contextual consensus representations. However, these methods suffer from a critical limitation: MVCC frameworks often regard similar heterogeneous views as positive sample pairs while treating dissimilar homogeneous views as negative sample pairs. This misalignment leads to intraclass inconsistency and interclass confusion. To address this problem, we propose a novel multiview clustering (MVC) method, termed semantic constraint-based spatial–spectral MVC (SCMVC). First, spatial views are designed to capture diverse features for contrastive clustering. Meanwhile, globally relevant information from the spectral view is extracted using a Transformer, which serves to enhance the representation of similar samples in the spatial multiview. Then, SCMVC employs a semantic constraint-based joint loss function, comprising a semantic contrast loss and a semantic similarity consistency loss. The semantic contrast loss captures high-level, domain-invariant features from HSIs, while the semantic similarity consistency loss enforces stricter constraints on the similarity of semantically related samples in feature space. Finally, SCMVC utilizes anchor points to guide similarity clustering, reducing randomness by predefining these points. This approach captures the directional characteristics of data, leading to a more stable clustering process. Abundant experimental studies on numerous benchmarks verify the superiority of SCMVC in comparison to some state-of-the-art clustering methods. The codes are available at https://github.com/YiLiu1999/SCMVC**

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***Index Terms*—Contrastive learning, hyperspectral image (HSI) clustering, multiview clustering (MVC).**

I. Introduction

HYPERSPECTRAL imaging is an advanced technology that integrates imaging and spectroscopic techniques [1], [2]. Unlike traditional color imaging, which captures only red, green, and blue bands, hyperspectral imaging records spectral information across hundreds of wavelengths, ranging from the ultraviolet to the near-infrared region. This process provides highly detailed spectral data, which is crucial for various applications, including precision agriculture [3] and urban planning [4]. Extensive research has been conducted on hyperspectral interpretation [5], [6], [7], [8]. High-precision interpretation typically relies on abundant, high-quality training samples with prior knowledge. However, acquiring large quantities of training samples is often labor-intensive, costly, and impractical. To address this challenge, unsupervised clustering has gained significant attention for its robust ability to reveal the intrinsic structure and hidden patterns within unlabeled hyperspectral image (HSI) data [9].

In particular, HSI clustering methods classify and label pixels based on their inherent similarity by maximizing intraclass dissimilarity and minimizing interclass similarity [10]. To map raw, complex data into a discriminative feature space for efficient clustering, many clustering methods focus on feature extraction techniques such as PCA, kernel methods, spectral methods, and deep neural networks [11]. Among these methods, deep learning-based clustering approaches leverage neural networks to extract high-level features from data, enabling the discovery of inherent structures and patterns. Notable examples include DCN [12], CCNN [13], and InfoGAN [14]. These methods have also achieved significant success in HSI clustering. Sun et al. [15] proposed a deep clustering network with an intraclass distance constraint for HSIs (DCIDC), enabling the network to simultaneously perform feature extraction and clustering. Lei et al. [16] introduced a deep spatial–spectral subspace clustering network (DS^{3}C-Net), which leverages multiscale auto-encoders and collaborative constraints to explore spatial–spectral information. Peng et al. [17] proposed a novel graph-based structural deep spatial–spectral clustering (GSDSSC) network to effectively explore the structural relationships among pixels. However, due to the complex spectral variations in HSI, these

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methods face challenges in learning pixel-level representations with sufficient discriminability for clustering.

As a powerful unsupervised learning paradigm, self-supervised learning (SSL) aims to learn representations that are beneficial for various downstream tasks [18], [19], without relying on prior knowledge. Owing to its ability to effectively extract meaningful features from high-dimensional, unlabeled hyperspectral data, SSL has been widely adopted in HSI clustering. Ding et al. [20] designed a self-supervised locality preserving low-pass graph convolutional embedding method (L2GCC), which extracts and clusters hyperpixel-level features in HSIs via spatial–spectral transformation and self-training strategies. Huang et al. [21] designed a self-supervised deep multilevel subspace clustering approach (S^2DMSC), which enables nonlinear representation and efficient clustering of large-scale HSIs. Mei and Yin [22] proposed a cascaded residual capsule network (CRCN) that leverages SSL with a coding rate reduction (CRR) mechanism and regularization to extract compact, transformation-invariant spatial–spectral features. However, existing SSL models are primarily designed for single-view HSIs, which limits their ability to fully exploit the rich and complementary information. The mainstream approach in SSL is dual-branch contrastive learning, where self-supervised models generate data representations by treating augmented views of the same sample as positive pairs. The training objective encourages positive pairs to be pulled closer in the feature space, while pushing negative pairs apart. For example, SimCLR [23] learns representations by maximizing the consistency between different augmented views of the same sample. BYOL [24] relies on interaction and mutual learning online. MoCo [25] facilitates unsupervised contrastive learning by constructing a dynamic dictionary with queues. SwAV [26] enforces consistency between cluster assignments from different views of the same image. Contrastive learning has also been widely used in HSI clustering [27], [28]. However, despite the promising results achieved by these methods, the collision problem [29] in contrastive learning remains unresolved. The collision problem refers to the increased probability of encountering negative samples with the same latent features as positive samples when the number of negative samples grows. These unexpected collisions hinder the model's ability to learn effective representations, leading to performance degradation. Meanwhile, a single neighbor pixel of HSI can make this problem more prominent due to extremely similar spectral and spatial features in its adjacent pixels. Such unintended collisions hinder the model's ability to learn discriminative representations and ultimately degrade performance.

To address this challenge in contrastive learning, several studies [30], [31], [32] have proposed integrating multiview clustering (MVC) with contrastive learning by leveraging the diverse and complementary information across different views. For example, Guan et al. [28] proposed a novel deep multiview subspace clustering algorithm (CMSCGC) to learn textural and spatial–spectral information. Zhao and Li [33] termed deep spectral clustering with regularized linear embedding (DSCRLE), to benefit from both spectral graph theory and deep learning techniques. Luo et al. [34] introduced a novel self-supervised double-structure transformer (SDST) to integrate the ground structure association information of HSI data. Tian et al. [35] proposed a correlation coefficient-based spectral delineation method (SSMLC) to generate multiple spectral views to obtain a consistent representation of HSI. Fang et al. [36] proposed a self-supervised multiview image clustering technique (GoMIC) under contrastive heterogeneous graph learning to reveal intraclass and interclass relationships of data. Hu et al. [37] designed a two-branch dense spectral–spatial network for HSI clustering to extract spectral and spatial features separately. However, existing multiview models primarily focus on ensuring consistency of the same samples across different views, while often overlooking the potential semantic consistency among distinct samples in the cross-view feature space.

In this study, we propose a novel semantic constraint-based spatial–spectral MVC (SCMVC) method to extract multiview consensus representations of HSI. SCMVC uses the spectral segmentation method to obtain multiview spatial features. Different from existing multiview constrative clustering methods that exploit the coherence goal of latent features to explore common semantics of all views. SCMVC considers the high similarity of information in neighboring heterogeneous spatial multiview data, which is difficult for general contrastive models to regard as negative sample pairs. First, the spatial multiview is designed to learn the diverse features for contrastive clustering, and the globally relevant information of the spectral view is extracted with Transformer [38] to enhance the spatial multiview differences between neighboring heterogeneous samples. Then, SCMVC employs a semantic constraint-based joint loss function, which consists of a semantic contrastive loss and a semantic similarity consistency loss. The semantic contrastive loss captures high-level domain-invariant features from HSI, while the semantic similarity consistency loss enforces tighter constraints on the similarity of semantically related samples in feature space. Finally, SCMVC employs an anchor-guide similarity clustering, which reduces randomness by pre-defining anchor points. This approach captures the directional characteristics of data, resulting in a more stable clustering process. Compared with the existing clustering methods, extensive experiments on four HSI datasets have demonstrated the effectiveness and the superiority of our proposed SCMVC. In summary, we highlight the main contributions of this article as follows.

1) We propose a novel SCMVC method, which utilizes spatial–spectral multiviews to reduce homogeneity between single views for extracting a multiview consensus representation of HSI.
2) We propose a semantic constraint-based joint loss to learn high-level domain-invariant features from HSI and enforce tighter constraints on the similarity of semantically related samples in feature space.
3) We employ an anchor-guided similarity clustering to reduce randomness by pre-defining anchor points, which capture the directional characteristics of data, resulting in a more stable clustering process.

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4) We qualitatively and quantitatively evaluate the performance of the proposed SCMVC model on four benchmark HSI datasets, which achieves superior performance in terms of land cover clustering.

The structure of this article is as follows. The related works of MVC and multiview contrastive clustering (MVCC) are reviewed in Section II. Section III describes the proposed SCMVC method in detail. Section IV conducts comprehensive experiments on four real HSI datasets to demonstrate the effectiveness of SCMVC.

II. Related Work

A. Multiview Clustering

MVC has attracted increasing attention due to its excellent performance in the unsupervised domain in recent years. MVC achieves feature extraction by fusing complementary and consensus information from each view. Existing MVC approaches mainly include four types: graph-based MVC, multiple kernel-based MVC, subspace-based MVC, and scalable-based MVC.

The graph-based MVC utilizes the topology of the graph and the connectivity between nodes to capture the intrinsic features and associative information of data to improve the clustering performance. Yang et al. [39] minimized the tensor Schatten p-norm to reduce the divergence between anchor graphs, which effectively preserves the manifold structure of each view. Nie et al. [40] proposed an auto-weighted multiple graph learning (AMGL) framework to learn a set of weights automatically for all the graphs. The multiple kernel-based MVC is able to capture the nonlinear structure inherent in many real-world data. Liu et al. [41] proposed a multiple kernel k-means (MKKMs) clustering algorithm with a matrix-induced regularization to reduce redundancy and enhance the diversity of the selected kernels. The subspace-based MVC assumes that data is usually located in underlying low-dimensional subspaces rather than evenly distributed throughout the space. Luo et al. [42] designed a subspace representation learning method that simultaneously learned a view-consistent representation and a set of view-specific representations for multiview subspace clustering. Zhang et al. [43] proposed a latent multiview subspace clustering (LMSC) method, which clusters data points with latent representation. The scalable-based MVC is a clustering method for processing large-scale datasets. Li et al. [44] proposed a scalable and parameter-free graph fusion framework for MVC. However, these MVC methods mainly focus on the consistency of the same samples across different views, often overlooking the case of similar but different samples in cross-view scenarios.

B. Multiview Contrastive Clustering

As one of the most effective self-supervised methods, contrastive learning aims to learn data representations by comparing the similarity between positive and negative samples. With the principle of pulling positive pairs close and pushing negative pairs apart, contrastive learning methods, such as SimCLR [23], MoCo [25], BYOL [24], and SwAV [26], have been widely studied in computer vision. Meanwhile, for the MVCC [45], the idea of contrastive learning is utilized to compare the similarity and difference between different views to improve data representation and clustering performance. CMC [46] found that the more views are introduced, the generated representation can reveal the underlying scene semantic information. MFLVC [30] introduced two objectives for MVC on high-level features and pseudo labels, employing contrastive learning. In this article, we propose a novel SCMVC method to extract multiview consensus representations of HSI.

III. Proposed Method

In this section, we introduce the SCMVC method, including the details of the multiview feature representation, the semantic constraint-based joint loss, and the anchor-guided similarity clustering. The overall framework of the proposed method is illustrated in Fig. 1.

A. Multiview Feature Representation

1) Spatial Multiview Construction: For multiview data, it is significant to make good use of the information from different views. Unlike natural images, HSIs have rich spectral information. In natural images, multiviews are obtained by data augmentation methods. In SCMVC, we use the spectral segmentation method to obtain spatial multiview data. Obviously, different bands in HSI can be considered as different views because they can reflect different spectral characteristics of the scene. To construct spatial multiview data $\{X_i\}_{i=1}^{v} \in \mathbb{R}^{p\times p\times k}$, where v is the number of views, p and k represent the patch size and the dimension after dimensionality reduction, respectively. We introduce the spatial view construction by principal component analysis (SvC-PCA) for HSI dimension reduction, as shown in Fig. 2. Particularly, the X_i from the original HSI $\mathcal{I} \in \mathbb{R}^{h\times w\times b}$, where h, w, and b represent the height, width, and band number, respectively. First, a sliding window is used to extract the pixel and its adjacent neighbor pixels, and the data points are represented with cubes $T \in \mathbb{R}^{p\times p\times b}$. Then, we split the T into v groups $\{G_i\}_{i=1}^{v} \in \mathbb{R}^{p\times p\times \lceil b/v \rceil}$, where $\lceil\rceil$ represents the ceiling function. We perform PCA [47] on each view to extract a low-dimensional spatial multiview data $\{X_i\}_{i=1}^{v}$. Finally, the spatial multiview data are transformed using multiple data augmentation methods (such as cropping, flipping, rotating, and shearing) to generate multiview features along the spatial dimension.

2) Spectral-Enhanced Spatial Multiview: To fully utilize the spatial–spectral information of HSI, we use a two-branch deep neural network. The network can map multiview samples into consensus representations for HSI clustering.

For the spatial view data $\{X^v\}_{v=1}^{V}$, which encodes the spatial information of each pixel position, we employ ResNet [48] as the backbone feature extractor to obtain spatial view representations. ResNet effectively captures the complex spatial structures inherent in the views. We denote the first and second spatial views of sample i as x_i^1 and x_i^2, respectively. In fact, the model ResNet is equivalent to a nonlinear embedding function $f_R(\cdot)$, which maps the spatial view inputs x_i^1 and x_i^2 into the embedding features hR_i^1 and hR_i^2. The formula is as follows:

$$hR_i^v = f_R\left(x_i^v\right) = ResNet\left(x_i^v\right) \tag{1}$$

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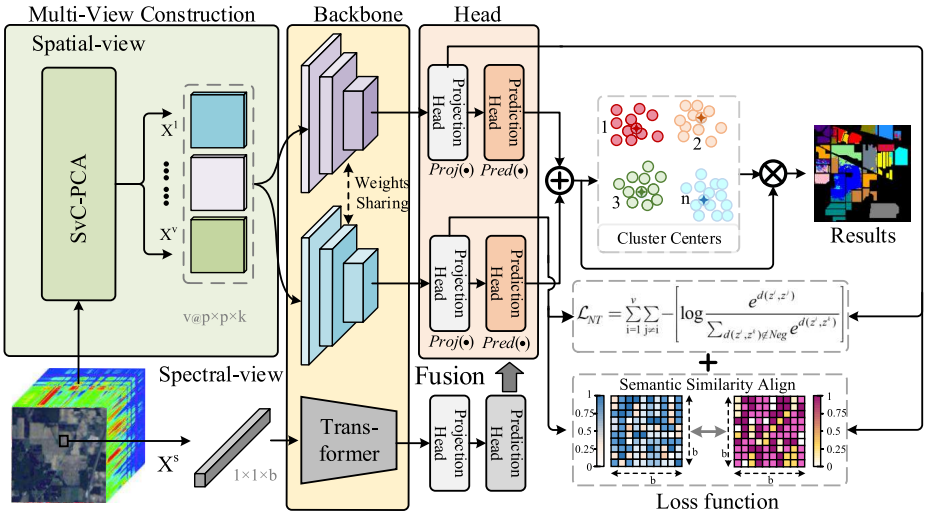


Fig. 1. Overall framework of the proposed SCMVC, which consists of three key components: multiview feature representation, semantic constraint-based joint loss, and anchor-guided similarity clustering. The multiview feature representation section maps the multiview data from the inputs as feature vectors. Subsequently, the semantic constraint-based joint loss is devised to mine potential consensus representation in the clustering space.

where $hR_i^v \in R^{N\times b}$ is the output after the average pooling layer. Then, we use a fully connected neural network with a single hidden layer as both the projection head $Proj(\cdot)$ and the prediction head $Pred(\cdot)$, and map the representation hR_i^v into the projection space and prediction space, respectively. The formula is as follows:

$$cR_i^v = Pred\left(hR_i^v\right) = W \cdot hR_i + b \tag{2}$$

$$zR_i^v = Proj\left(cR_i^v\right) = Norm\left(W \cdot \sigma\left(cR_i^v\right) + b\right) \tag{3}$$

where σ denotes the ReLU activation function, $Norm(x) = x_i / \left(\left(\sum_j x_j^2\right)^{1/2}\right)$ represents a normalization operation, and the W and b are the weight and bias parameters of the hidden layer.

Considering the spectral data of each pixel of the HSI changes continuously, the global relationship is very important to understand the entire spectral data. Therefore, we use a Transformer to extract globally relevant information from the spectral data. We use $x_i^s \in R^{1\times1\times b}$ to represent the ith sample in the spectral dimension, where b is the dimension of the spectral data. The standard Transformer architecture is composed of multihead self-attention modules, residual connections, and multilayer perceptron (MLP) blocks. Each block is preceded by a layer normalization operation, and residual connections are integrated after each block. The self-attention mechanism enables the Transformer to model global relationships among input embeddings, thereby enhancing its ability to capture long-range spectral dependencies in HSI data. The Transformer model is also equivalent to a nonlinear embedding function $f_T(\cdot)$, which maps the spectral data $x_i^s \in R^{1\times1\times b}$ into the embedding features hT_i. The function is as follows:

$$hT_i = f_T\left(x_i^s\right) = Transformer\left(x_i^s\right). \tag{4}$$

For HSI, in each pixel, adjacent pixels of different spectral channels are strongly correlated, and adjacent pixels often belong to the same object (especially for very high-resolution data). However, adjacent pixels that are not in the same category also have similar spatial characteristics. General contrastive loss directly utilizes the data representation obtained by these spatial features, and these samples will be regarded as false negative pairs. In order to enhance the similarity between the positive samples and maximize the difference between the negative samples, we use the extracted data representation hT_i from the spectral model to strengthen the extracted data representation cR_i^v and zR_i^v from the spatial model. In order to retain sufficient categorization information in the prediction space, we fuse the label prediction probabilities of multiview, expressed as follows:

$$h = \sum_{i=1}^{v} Norm\left(cR^i\right) + Norm\left(pred\left(hT_i\right)\right) \tag{5}$$

where cR^i denotes the ith spatial view and $pred(\cdot)$ denotes the prediction head used after the Transformer block. The consensus representation z_i^v is expressed as follows:

$$z_i^v = zR_i^v + Norm\left(proj\left(pred\left(hT_i\right)\right)\right) \tag{6}$$

where $proj(\cdot)$ denotes the projection head used after the Transformer block. Therefore, the obtained consensus representation features encompass the local spatial structure relationships from the spatial view and the long-range dependency relationships from the spectral view.

B. Semantic Constraint-Based Joint Loss

For clustering, we design a semantic constraint joint loss to effectively integrate the two-branch deep neural network in a unified training framework. The overall loss function is formulated as follows:

$$\mathcal{L} = \mathcal{L}_{SC} + \alpha\mathcal{L}_{SSC} \tag{7}$$

where $\alpha \geq 0$ is a hyperparameter that balances the contribution of semantic contrastive loss $\mathcal{L}_{SC}$ and semantic similarity consistency loss $\mathcal{L}_{SSC}$.

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1) Semantic Contrastive Loss: Contrastive learning can learn representations from unlabeled data. It can sample positive pairs, which should be similar samples with similar representations. We use normalized temperature-scaled cross entropy loss [23], which learns discriminative representations by maximizing consistency between different augmented views of the same instance via a contrastive loss in the latent space. Specifically, each positive pair in our network is formed by applying two different augmentations to the same sample, resulting in representations for contrastive training. The objective of contrastive learning is to maximize the similarity between positive pairs while minimizing the similarity between negative pairs. Then, the similarity between two features can be measured by the cosine distance as follows:

$$d\left(z^i, z^j\right) = \frac{\langle z^i, z^j \rangle}{\left\| z^i \right\| \left\| z^j \right\|} \tag{8}$$

where $\langle \cdot, \cdot \rangle$ represents the dot product operator. And then, the feature consistency objective $\mathcal{L}_{SC}$ can be formulated as

$$\mathcal{L}_{SC} = \sum_{i=1}^{v} \sum_{j \neq i} - \left[\log \frac{e^{d(z^i, z^j)/\tau}}{\sum_{d(z^i, z^k) \notin Neg} e^{d(z^i, z^k)/\tau}} \right] \tag{9}$$

where *Neg* denotes negative feature pairs and $d(z^i, z^k)$ is the similarity of negative feature pairs. τ is the temperature parameter.

2) Semantic Similarity Consistency Loss: In HSI data clustering, semantic contrastive loss enhances clustering performance by clustering features of similar samples together and pushing features of dissimilar samples apart. However, relying solely on semantic contrastive loss might be insufficient for capturing the global semantic relationships among feature representations. Therefore, we design a semantic similarity consistency loss function to further align the semantic similarities of different feature representations, thereby enhancing the effect of semantic contrastive loss. Given feature representation matrix z^v, we first compute the similarity matrix $\mathbf{S}^v$

$$\mathbf{S}^v = softmax\left(\frac{\mathbf{z}^v \mathbf{z}^{v\top}}{n}\right) \tag{10}$$

where $\mathbf{z}^v$ denotes the feature representation matrix of the vth view and n is the number of samples. The similarity matrix $\mathbf{S}^v$, computed by dot products, captures the pairwise similarities between the features in $\mathbf{z}^v$. A softmax operation is subsequently applied to normalize the similarity values and ensure they lie within a comparable range. Finally, the mean squared error (mse) is employed to measure the discrepancy between similarity matrices

$$\mathcal{L}_{SSC} = \sum_{i=1}^{v} \sum_{j \neq i} \left(\mathbf{S}^i - \mathbf{S}^j\right)^2 \tag{11}$$

where $\mathcal{L}_{SSC}$ denotes the semantic similarity consistency loss, which quantifies the discrepancy between similarity matrices $\mathbf{S}^i$ and $\mathbf{S}^j$ by summing the squared differences of their corresponding elements. $\mathcal{L}_{SSC}$ ensures the consistency of different feature representations in the similarity space by aligning their semantic similarities. By promoting such consistency, $\mathcal{L}_{SSC}$ enhances the model's ability to capture the intrinsic structure of HSI data and improves the robustness of the learned representations. When combined with contrastive loss, it significantly boosts the clustering performance of HSI data.

C. Anchor-Guided Similarity Clustering

In this section, we describe the procedure for obtaining the final clustering labels. Given the input feature matrix $h \in \mathcal{R}^{N \times d}$ from (5) and an anchor center matrix $A \in \mathcal{R}^{K \times d}$, we first compute the cosine similarity between each sample and the anchor centers using the following formula:

$$C = Norm\left(h \cdot A^{\top}\right) \tag{12}$$

where $C \in \mathcal{R}^{N \times K}$ represents the similarity between samples and anchor centers. Each row of the matrix is normalized to ensure that the resulting values represent cosine similarity scores. The similarity matrix C is processed through a softmax function to produce a probability distribution for each sample over all clusters

$$p_{ij} = \frac{e^{C_{ij}}}{\sum_{k=1}^{K} e^{\mathbf{C}_{ik}}} \tag{13}$$

where C_{ij} represents the probability that the ith sample belongs to the jth anchor center. We assign the initial clustering label to each sample by selecting the index of the highest probability

$$\hat{y}_i = \arg\max_j \left(p_{ij}\right), \quad i = 1, 2, \ldots, N. \tag{14}$$

Next, we apply the K-means clustering algorithm to the feature matrix h. The anchor center matrix can be updated by minimizing the sum of squared distances between each sample and its assigned clustering center

$$\hat{A} = \min_{A_1, \ldots, A_K} \sum_{i=1}^{N} \sum_{j=1}^{K} r_{ij} \left\| h_i - A_j \right\|^2 \tag{15}$$

where A_j is the jth cluster center. r_{ij} is a binary assignment indicator that specifies whether sample h_i is assigned to cluster j, which is defined as

$$r_{ij} = \begin{cases} 1, & \text{if } h_i \text{ is assigned to cluster } j \\ 0, & \text{otherwise.} \end{cases} \tag{16}$$

Finally, to evaluate the performance of the clustering method, we use a matching function $y_i = Hungarian(gt_i, \hat{y}_i)$ to align the predicted cluster labels $\hat{y}_i$ with the ground truth labels gt_i. This matching is performed by using the Hungarian algorithm [49].

IV. Experimental Results and Analysis

In this section, four benchmark datasets are employed to evaluate the effectiveness of the proposed SCMVC method. The experimental results are then thoroughly analyzed and discussed in detail.

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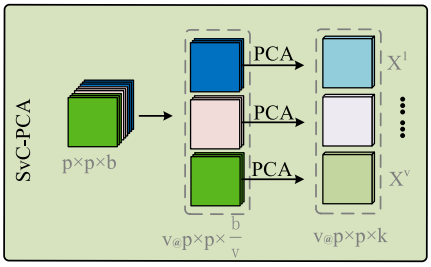


Fig. 2. Spatial view construction by principal component analysis.

A. Experiment Setup

SCMVC was implemented in Python (version 3.10.6) using the PyTorch framework. All experiments were conducted on a workstation equipped with an NVIDIA RTX A6000 GPU (48 GB). The model was trained for 200 epochs until convergence by minimizing the semantic constraint-based joint loss defined in (7), using the Adam optimizer [50]. ResNet-50 and Transformer were adopted as the default backbones for the spatial and spectral models, respectively. The projection head, $Proj(\cdot)$, is implemented as an MLP with a hidden dimension of 128, and its output is not followed by batch normalization (BN) or ReLU activation. The prediction head, $Pred(\cdot)$, is also an MLP with a hidden dimension of 2048, and its output includes a ReLU activation. The batch size is set to 128, and the temperature parameter in the contrastive loss is fixed at 0.5.

We conducted experiments on multiple benchmark datasets to compare the following state-of-the-art clustering algorithms: MVC algorithms such as MFLVC (CVPR 2022) [30], GCFAGG (CVPR 2023) [51], ACCMVC (TNNLS 2024) [52], and CMSCGC (TGRS 2024) [28]; traditional HSI clustering algorithms like DGAE (TCSVT 2022) [53], GCOT (TGRS 2022) [54], EGCSC (TGRS 2020) [55], and EKGCSC (TGRS 2020) [55]; and deep-based HSI clustering algorithms including MS^{2}A-Net(JSTARS 2022) [56], DSCRLE (TGRS 2023) [33], and SDST (TGRS 2024) [34].

The performance of various clustering methods is quantitatively evaluated using eight popular clustering metrics. These metrics include overall accuracy (ACC), each class accuracy, kappa coefficient (Kappa), clustering purity (Purity), adjusted rand index (ARI), adjusted mutual information (AMI), Fowlkes–Mallows index (FMI), and normalized mutual information (NMI). Furthermore, we use the Hungarian algorithm to optimally match the predicted cluster labels with the ground truth labels, ensuring accurate assignments before computing the evaluation metrics. For all metrics, the higher the value is, the better the clustering performance is.

B. Dataset Description

In our experimental analysis, the effectiveness of the proposed SCMVC is evaluated by using four real HSIs, including Salinas, Botswana, Houston 2013, and Indian-Pines. These datasets are used to comprehensively assess the clustering performance of our proposed method in Table I.

1) Salinas: The first dataset (Salinas) was collected using a 224-band airborne visible/infrared imaging spectrometer (AVIRIS) sensor over the Salinas Valley, in California, USA. With a high spatial resolution (3.7 m), this dataset covers an area of 512 rows and 217 columns. It contains 224 bands with a spectral wavelength from 0.4 to 2.5 μm. In addition, some unannotated man-made materials have been annotated as unknown samples, and the reference map contains 16 classes in total. The number of each class is presented in Fig. 3, and the pseudo-color and the corresponding ground truth map are shown in Fig. 4(a).

TABLE I
STATISTICS SUMMARY ON THE FOUR HSI DATASETS

Dataset	Clusters	Samples	Views	Patch-size	Bands
Salinas	16	54129	3	28×28×3	224
Botswana	14	3248	3	28×28×3	145
Houston 2013	15	15029	3	28×28×3	144
Indian-Pines	16	10249	3	28×28×3	200

2) Botswana: The second dataset (Botswana), sourced from the EO-1 satellite of NASA, was collected between 2001 and 2004 over the Botswana region. The image has 242 bands with a pixel resolution of 30 m. These bands span the range of 400–2500 nm at 10-nm intervals. After eliminating noise bands, 145 candidate features were retained, including [10–55, 82–97, 102–119, 134–164, and 187–220]. The dataset comprises 14 classes. The number of each class is presented in Fig. 3, and the pseudo-color and the corresponding ground truth map are shown in Fig. 4(b).

3) Houston 2013: The third dataset (Houston 2013) was acquired over the University of Houston campus and the neighboring urban area. It was released for the IEEE GRSS DFC 2013. It consists of 144 spectral bands in the 380–1050 nm region with 1905 × 349 pixels. Its spatial resolution is 2.5 m. There are 15 distinguishable land cover categories investigated in the scene. The number of each class is presented in Fig. 3, and the pseudo-color and the corresponding ground truth map are shown in Fig. 4(c).

4) Indian-Pines: The fourth dataset (Indian-Pines), acquired by the AVIRIS sensor, contains the Indian-Pines test site in Northwestern Indiana. After excluding water absorption bands, this image encompasses 200 spectral bands with 145 × 145 pixels. Its spectral coverage spans from 0.4 to 2.5 μm, with a spatial resolution of 20 m. There are two major dual-lane highways, a rail line, as well as some low-density housing, other built structures, and smaller roads. Since the scene is taken in June, some of the crops present, corn, soybeans, are in early stages of growth with less than 5% coverage. The ground truth available is designated into 16 classes and is not all mutually exclusive. We have also reduced the number of bands to 200 by removing bands covering the region of water absorption: [104–108], [150–163], and 220. The number of each class is presented in Fig. 3, and the pseudo-color and the corresponding ground truth map are shown in Fig. 4(d).

C. Clustering Results Analysis

In this section, we present the clustering results of the proposed SCMVC method along with several state-of-the-art

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Salinas		Botswana		Houston 2013		Indian-Pines	
Class Name	Number	Class Name	Number	Class Name	Number	Class Name	Number
Brocoli-green-weeds1	2009	Water	270	Healthy-grass	1251	Alfalfa	46
Brocoli-green-weeds2	3726	Hippo-grass	101	Stressed-grass	1254	Corn-notill	1428
Fallow	1976	FloodPlain-grasses1	251	Synthetic-grass	697	Corn-mintill	830
Fallow-rough-plow	1394	FloodPlain-grasses2	215	Tress	1244	Corn	237
Fallow-smooth	2678	Reeds	269	Soil	1242	Grass-pasture	483
Stubble	3959	Riparian	269	Water	325	Grass-trees	730
Celery	3579	Firescar	259	Residential	1268	Grass-pasture-mowed	28
Grapes-untrained	11271	Island-interior	203	Commercial	1244	Hay-windrowed	478
Soil-vinyard-develop	6203	Acacia-woodlands	314	Road	1252	Oats	20
Corn-senesced-green-weeds	3278	Acacia-shrublands	248	Highway	1227	Soybean-notill	972
Lettuce-romaine-4wk	1068	Acacia-grasslands	305	Railway	1235	Soybean-mintill	2455
Lettuce-romaine-5wk	1927	Short-mopane	181	Parking-Lot1	1233	Soybean-clean	593
Lettuce-romaine-6wk	916	Mixed-mopane	268	Parking-Lot2	469	Wheat	205
Lettuce-romaine-7wk	1070	Chalcedony	95	Tennis-Court	428	Woods	1256
Vinyard-untrained	7268			Running-Track	660	Buildings-Grass-Trees-Drivers	386
Vinyard-vertical-trellis	1807					Stone-Steel-Towers	93
Total	54129	Total	3248	Total	15029	Total	10249

Fig. 3. Land-cover types of the Salinas, Botswana, Houston 2013, and Indian-Pines dataset.

TABLE II

CLUSTERING PERFORMANCE % OF THE PROPOSED SCMVC COMPARED TO THE STATE-OF-THE-ART COMPETITORS ON SALINAS. THE BEST AND SECOND-BEST VALUES OF THE CLUSTERING RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Class	MFLVC	GCFAGG	ACCMVC	GCOT	EGCSC	EKGCSC	MS^{2}A-Net	DSCRLE	SDST	SCMVC
Brocoli-green-weeds1	84.85	27.50	90.83	0.05	0.00	91.94	0.00	95.12	0.00	**100.00**
Brocoli-green-weeds2	81.59	52.97	93.55	93.16	**100.00**	0.00	99.09	99.57	99.89	99.95
Fallow	94.81	36.50	**96.00**	58.15	0.00	0.00	69.43	57.89	35.22	90.23
Fallow-rough-plow	98.49	76.85	95.04	96.99	0.00	**100.00**	99.35	95.41	94.40	99.35
Fallow-smooth	82.68	83.88	89.30	71.66	**100.00**	98.06	46.83	88.76	96.49	98.54
Stubble	88.50	87.88	92.72	62.79	**100.00**	78.93	99.49	99.49	98.59	99.67
Celery	94.24	28.72	95.97	87.01	0.00	91.67	0.00	98.74	**99.22**	98.55
Grapes-untrained	55.65	53.30	45.92	96.82	83.08	85.23	94.69	**97.91**	63.31	86.62
Soil-vinyard-develop	95.12	68.74	96.46	91.17	98.03	96.37	93.60	**99.11**	97.40	75.09
Corn-senesced-green-weeds	91.90	45.64	89.96	94.20	**97.90**	97.74	34.75	65.80	65.01	86.15
Lettuce-romaine-4wk	**93.81**	8.43	5.44	0.00	0.00	0.00	39.61	2.15	0.00	0.00
Lettuce-romaine-5wk	90.42	74.01	86.43	**100.00**	0.00	98.96	46.34	98.03	**100.00**	97.25
Lettuce-romaine-6wk	0.00	86.01	0.00	0.00	0.00	0.00	96.07	97.82	**98.58**	78.28
Lettuce-romaine-7wk	93.64	6.18	91.02	92.15	0.00	0.00	0.00	88.22	88.22	**94.39**
Vinyard-untrained	33.70	19.42	46.40	0.00	7.65	9.04	2.39	3.47	66.95	**71.88**
Vinyard-vertical-trellis	93.95	31.67	84.75	**99.67**	0.00	0.00	3.93	70.50	72.27	98.06
ARI	57.40	37.29	60.10	61.33	49.07	57.40	51.09	67.24	60.50	**74.83**
AMI	75.61	53.14	76.76	81.34	72.00	74.12	69.01	85.09	78.37	**87.09**
FMI	61.71	43.29	64.35	68.48	58.67	63.65	60.96	73.32	64.56	**77.37**
NMI	75.63	53.18	76.78	81.36	72.03	74.14	69.04	85.11	78.39	**87.10**
Purity	78.26	56.04	82.32	73.90	57.11	64.28	59.74	80.14	76.09	**89.41**
Kappa	71.80	45.24	71.09	66.00	48.71	57.59	52.95	75.85	73.12	**85.08**
ACC	74.20	50.13	73.41	69.95	54.63	62.12	58.56	78.63	75.80	**86.56**

algorithms across four benchmark datasets. The evaluation metrics are reported in separate tables to quantitatively compare the performance of different clustering approaches. Each row in the tables corresponds to the results of different clustering methods under the same experimental settings. For fair comparison, MVC methods such as MFLVC [30], GCFAGG [51], and ACCMVC [52] are evaluated using the same multiview inputs as SCMVC. In the case of multiview subspace clustering methods like CMSCGC [28], spatial and spectral features are used as distinct input views. To further illustrate the clustering effectiveness, we also provide visual comparisons of clustering maps in the subsequent figures.

1) Quantitative Evaluation: From Tables II to V, it can be observed that our proposed SCMVC method consistently delivers significant performance improvements while maintaining competitive computational efficiency. Across all datasets, the performance of our proposed method always ranks in the top two with respect to the seven comprehensive clustering criteria: ACC, Kappa, NMI, Purity, ARI, AMI, FMI, and NMI. Specifically, in terms of ACC, SCMVC improves upon DSCRLE by 7.93%, 1.79%, and 13.94% on the Salinas, Botswana, and Houston 2013 datasets, respectively. On the Indian-Pines dataset, SCMVC also surpasses EKGCSC by 4.06%. Furthermore, for large-scale datasets such as Houston 2013, several methods encounter memory errors during clustering, including GCOT, EGCSC, and EKGCSC. Similarly, when the number of samples is very large (e.g., Salinas with 54 129 samples), the multiview subspace clustering method CMSCGC also experiences a memory error. In contrast, SCMVC not only achieves high-precision clustering but also

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TABLE III

CLUSTERING PERFORMANCE % OF THE PROPOSED SCMVC COMPARED TO THE STATE-OF-THE-ART COMPETITORS ON BOTSWANA. THE BEST AND SECOND-BEST VALUES OF THE CLUSTERING RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Class	MFLVC	GCFAGG	ACCMVC	CMSCGC	DGAE	GCOT	EKGCSC	MS^{2}A-Net	DSCRLE	SDST	SCMVC
Water	57.65	53.10	67.55	**100.00**	65.18	99.63	**100.00**	**100.00**	**100.00**	98.15	**100.00**
Hippo-grass	93.41	63.92	93.81	**100.00**	92.08	0.00	0.00	0.00	**100.00**	88.12	**100.00**
FloodPlain-grasses1	84.32	51.50	85.48	96.41	84.06	66.53	94.42	98.80	99.60	0.80	**100.00**
FloodPlain-grasses2	64.39	48.04	57.08	93.49	73.49	82.3	**100.00**	97.67	**100.00**	0.00	**100.00**
Reeds	69.11	57.03	62.88	72.86	64.31	75.46	13.01	40.15	67.66	75.46	**84.01**
Riparian	40.15	31.08	44.36	62.45	0.00	0.00	0.00	0.37	53.90	**88.85**	68.03
Firescar	82.38	75.43	93.31	99.23	82.63	0.00	**100.00**	0.00	99.23	92.28	**100.00**
Island-interior	66.67	79.17	58.08	71.92	0.00	71.92	71.92	21.18	98.52	72.41	**100.00**
Acacia-woodlands	63.27	46.26	58.71	88.85	69.43	**100.00**	**100.00**	69.43	92.04	63.38	97.77
Acacia-shrublands	98.31	92.02	96.34	91.94	51.21	0.00	0.00	77.82	99.60	94.76	**100.00**
Acacia-grasslands	64.48	33.33	75.25	72.46	90.49	**100.00**	**100.00**	94.10	95.08	93.11	76.39
Short-mopane	85.96	64.53	87.22	96.13	82.32	4.97	3.87	54.13	98.90	81.77	**100.00**
Mixed-mopane	93.57	77.52	86.01	98.88	**100.00**	0.00	**100.00**	67.91	**100.00**	94.78	**100.00**
Chalcedony	12.09	70.97	0.00	81.05	80.00	16.84	60.00	52.11	83.16	83.16	**89.47**
ARI	55.99	40.45	56.94	75.14	51.20	41.95	56.59	49.58	**90.98**	60.60	86.50
AMI	69.21	54.84	68.62	85.31	70.48	71.29	82.96	68.48	87.22	75.46	**91.93**
FMI	59.34	44.95	60.21	77.12	55.68	54.13	65.23	54.69	**90.89**	64.32	87.54
NMI	69.54	55.33	68.94	85.89	70.79	71.63	83.15	68.81	86.17	75.73	**92.01**
Purity	73.40	62.11	73.50	86.98	69.30	51.08	66.26	61.67	91.50	74.11	**93.29**
Kappa	68.39	55.34	68.16	85.46	62.99	44.67	61.82	56.13	90.80	71.09	**92.73**
ACC	70.67	58.53	70.47	86.98	65.86	49.45	65.06	59.61	91.50	73.37	**93.29**

TABLE IV

CLUSTERING PERFORMANCE % OF THE PROPOSED SCMVC COMPARED TO THE STATE-OF-THE-ART COMPETITORS ON HOUSTON 2013. THE BEST AND SECOND-BEST VALUES OF THE CLUSTERING RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Class	MFLVC	GCFAGG	ACCMVC	CMSCGC	MS^{2}A-Net	DGAE	DSCRLE	SDST	SCMVC
Healthy-grass	65.04	44.63	63.49	71.30	**85.21**	41.67	84.81	72.82	84.89
Stressed-grass	16.26	11.27	13.83	53.75	39.07	10.05	**83.57**	37.24	66.51
Synthetic-grass	64.08	10.51	84.48	**99.28**	0.00	**100.00**	94.98	99.14	99.14
Tress	49.63	31.25	54.64	75.56	75.64	14.63	92.85	52.33	**94.13**
Soil	60.64	51.86	58.34	78.74	87.44	47.67	82.05	83.66	**100.00**
Water	46.13	29.25	72.00	**77.23**	1.23	0.00	61.85	51.69	76.62
Residential	36.10	21.05	51.39	59.31	54.10	18.22	54.81	54.89	**79.26**
Commercial	20.83	19.02	24.35	8.84	13.50	15.03	22.99	14.39	**31.75**
Road	40.79	21.77	59.29	79.47	21.49	10.06	73.24	73.88	**91.77**
Highway	31.72	20.17	32.49	26.49	**41.32**	39.85	26.57	26.57	36.39
Railway	52.49	32.65	50.00	56.28	5.67	83.08	62.27	67.94	**88.10**
Parking-Lot1	23.17	17.35	23.03	21.90	**53.52**	30.49	27.58	23.84	46.72
Parking-Lot2	9.27	11.69	4.51	0	8.96	69.30	56.50	0.00	**95.74**
Tennis-Court	79.58	1.43	90.89	98.83	0.00	**100.00**	0.00	96.73	**100.00**
Running-Track	61.38	36.07	39.51	99.85	12.58	0.00	98.12	99.24	**100.00**
ARI	23.44	10.19	26.59	39.16	26.31	21.11	47.11	33.18	**64.40**
AMI	39.40	20.01	41.28	60.41	46.19	39.94	64.14	53.59	**77.17**
FMI	29.16	16.82	31.99	44.19	34.25	27.70	51.27	38.93	**67.04**
NMI	39.55	20.21	41.43	54.34	46.33	40.10	64.23	53.71	**77.23**
Purity	44.71	28.54	49.31	58.07	41.89	39.50	64.60	55.15	**77.24**
Kappa	37.58	20.00	41.51	60.51	35.17	30.63	59.50	51.33	**74.60**
ACC	42.08	25.56	45.69	57.60	40.40	35.27	62.49	54.93	**76.43**

significantly outperforms these methods in terms of large-scale data.

By analyzing the clustering accuracy for each category, we observe that certain classes are not recognized by clustering methods, leading to extremely low or even zero accuracy (i.e., ACC = 0). For example, the "Lettuce-romaine-6wk" class in the Salinas dataset, the "Chalcedony" class in the Botswana dataset, the "Parking-Lot2" class in the Houston 2013 dataset, and the "Oats" class in the Indian-Pines dataset exhibit this issue. As shown in Fig. 4, these classes have 469, 95, 916, and 20 samples, respectively, which are the smallest classes in the datasets. For HSI clustering, these smaller classes are prone to misclassification due to the dominance of larger classes. Our proposed SCMVC method addresses this problem through MVC and a semantic constraint-based joint loss function. As a result, SCMVC tends to produce fewer empty clusters compared to the other methods.

Furthermore, as shown in Tables II–V, most MVC methods outperform GCOT on the four datasets. While GCOT only relies on the geometric structure of features extracted

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TABLE V

CLUSTERING PERFORMANCE % OF THE PROPOSED SCMVC COMPARED TO THE STATE-OF-THE-ART COMPETITORS ON INDIAN-PINES. THE BEST AND SECOND-BEST VALUES OF THE CLUSTERING RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Class	MFLVC	GCFAGG	ACCMVC	CMSCGC	DGAE	GCOT	EGCSC	EKGCSC	MS^{2}A-Net	DSCRLE	SDST	SCMVC
Alfalfa	0.00	52.17	0.00	82.61	97.83	0.00	0.00	0.00	0.00	0.00	0.00	**100.00**
Corn-notill	40.67	41.68	39.97	26.26	49.09	35.50	46.22	**64.78**	30.60	18.84	41.11	44.33
Corn-mintill	22.89	33.90	37.88	33.25	31.45	8.55	18.31	27.59	23.98	29.76	**40.60**	40.48
Corn	97.05	31.65	98.31	44.73	0.00	0.00	62.03	0.00	16.88	25.74	14.35	**99.58**
Grass-pasture	56.52	27.12	**65.35**	61.70	48.65	0.41	0.00	12.22	6.21	52.80	52.17	63.35
Grass-trees	60.82	39.64	61.23	90.27	35.07	88.49	94.93	63.01	61.37	87.40	79.73	**95.21**
Grass-pasture-mowed	0.00	3.57	0.00	0.00	**100.00**	0.00	**100.00**	**100.00**	0.00	**100.00**	3.57	00.00
Hay-windrowed	96.02	23.48	97.91	55.86	**100.00**	**100.00**	**100.00**	**100.00**	89.75	58.16	99.16	**100.00**
Oats	40.00	70.00	0.00	0.00	0.00	0.00	45.00	**100.00**	65.00	0.00	35.00	5.00
Soybean-notill	45.20	21.19	**73.33**	44.86	59.88	32.10	3.70	14.92	31.58	48.05	20.16	70.47
Soybean-mintill	46.70	30.41	34.58	42.00	76.13	71.24	**81.67**	73.40	46.84	49.82	73.48	56.37
Soybean-clean	44.52	44.86	70.10	43.00	72.18	2.02	50.76	**88.20**	16.19	20.40	13.32	68.13
Wheat	99.02	0.49	98.54	**100.00**	99.51	60.00	58.54	**100.00**	65.37	93.17	96.59	**100.00**
Woods	71.12	73.26	70.83	38.81	46.72	89.01	**90.43**	87.43	92.57	59.84	63.00	71.46
Buildings-Grass-Trees-Drivers	75.06	61.92	76.17	37.56	75.13	8.55	43.52	**76.94**	10.10	12.44	30.83	**76.94**
Stone-Steel-Towers	24.73	1.08	0.00	90.32	0.00	**100.00**	0.00	0.00	0.00	98.92	96.77	89.25
ARI	36.52	22.05	37.84	27.90	48.99	33.26	38.96	**49.69**	29.04	48.79	34.63	48.41
AMI	56.28	37.05	61.28	51.86	65.25	51.57	62.84	**68.35**	41.16	37.59	51.44	66.49
FMI	43.81	30.24	45.25	35.70	54.90	43.65	50.32	**56.10**	37.08	48.55	43.45	54.78
NMI	56.48	37.33	61.45	40.46	65.41	51.81	63.03	**68.50**	41.43	29.90	51.68	66.64
Purity	64.73	50.27	70.51	58.34	72.52	54.75	64.30	69.89	50.51	56.62	57.62	**75.52**
Kappa	49.32	32.13	52.43	52.09	53.51	42.42	50.85	56.10	37.29	40.22	47.49	**62.06**
ACC	53.17	38.14	55.80	45.53	58.23	50.27	57.97	61.24	43.84	45.61	54.22	**65.30**

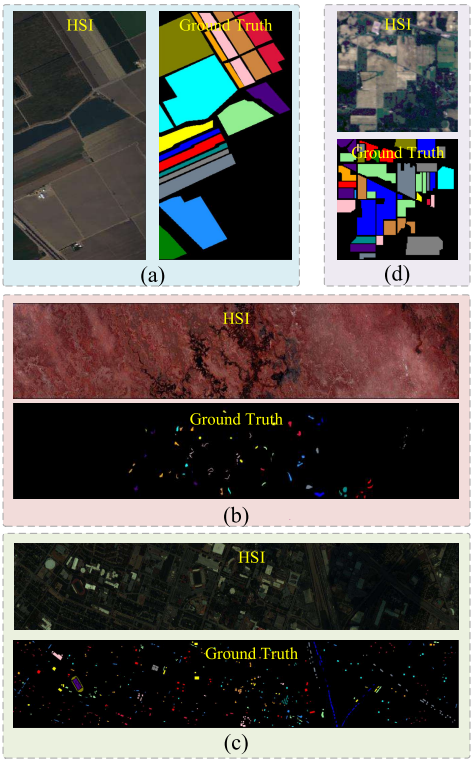


Fig. 4. False-color visualizations and ground truth for four datasets. Black denotes unknown classes for each ground truth image. (a) Salinas dataset. (b) Botswana dataset. (c) Houston 2013 dataset. (d) Indian-Pines dataset.

from the original data, MVC leverages multiple views to aggregate and complement high-level semantic features. Both EGCSC and EKGCSC perform clustering under the GCSC framework. However, the GCSC framework suffers from quadratic complexity concerning the size of data points, making it impractical for large-scale HSI data (such as Botswana and Houston 2013). Among deep learning-based methods, DSCRLE, which uses orthogonally constrained latent features, outperforms the dual-branch network (i.e., SDST) in most conditions, while the proposed network achieves the best results among these networks. The proposed SCMVC method demonstrates more advantages in different scenarios, as it not only utilizes multiview data to extract complementary information but also fully leverages semantic loss to align data samples. Specifically, for the large-scale HSI datasets (i.e., Houston 2013 and Botswana), SCMVC achieves a performance improvement over the best comparison method (i.e., DSCRLE) by 13.94% and 1.79%, respectively. Similarly, on the Salinas and Indian-Pines datasets, SCMVC outperforms the best comparison methods (i.e., DSCRLE and EKGCSC) by 7.93% and 4.06%, respectively. Overall, the proposed SCMVC achieves superior results with the other comparison methods. The reason why our SCMVC outperforms the other methods is that the spatial–spectral MVC scheme and the semantic-contrastive self-supervised loss are introduced in the learning model.

2) Qualitative Evaluation: Figs. 5–8 present the visual clustering maps of different methods on the four HSI datasets. From the figures, it can be observed that, compared with all the other benchmarked methods, the land-cover regions of the same category acquired by our proposed SCMVC are more uniform with the fewest misclassified pixels. This is particularly evident in the areas where the intermixed ground objects have similar spectral characteristics. Taking the Salinas dataset as an example (see Fig. 5), the pixels for the land cover types of "Corn-notill," "Corn-mintil," and "Corn" are prone to being misclassified by most methods due to their overly similar spectral features. Most notably, our method can effectively distinguish these kinds of land cover types. Our method adopts an advanced multiview feature extraction method to effectively identify and exploit spatial and spectral features in HSI data. This is especially evident when processing the Botswana dataset, which enables a clearer distinction between water and nonwater areas. By comparing clustering maps, we demonstrate the difference between our method and the traditional

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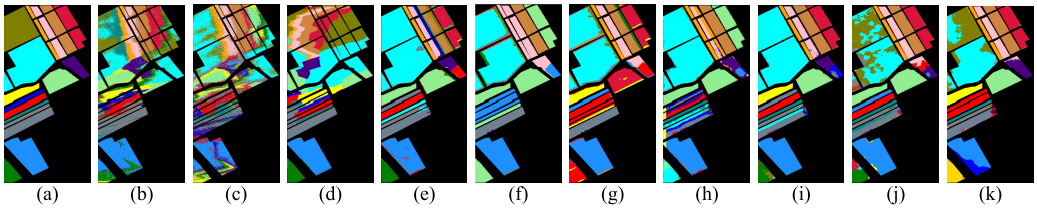


Fig. 5. Clustering maps of the Salinas image. (a) Ground truth. (b) MFLVC. (c) GCFAGG. (d) ACCMVC. (e) GCOT. (f) EGCSC. (g) EKGCSC. (h) MS^{2}A-Net5. (i) DSCRLE. (j) SDST. (k) SCMVC.

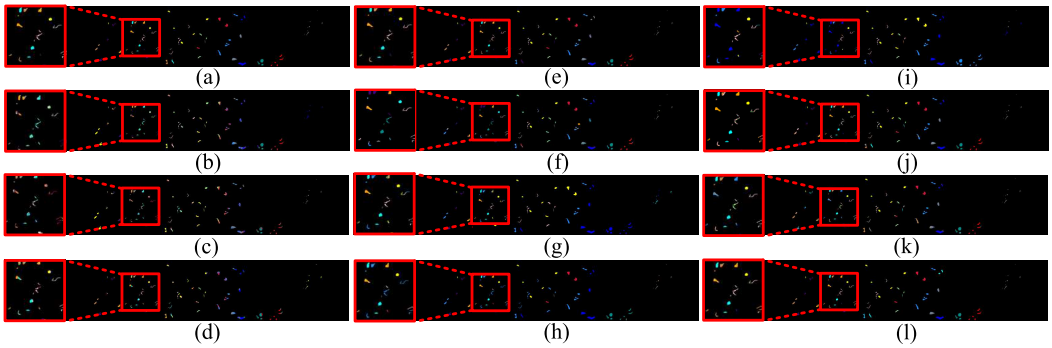


Fig. 6. Clustering maps of Botswana image. (a) Ground truth. (b) MFLVC. (c) GCFAGG. (d) ACCMVC. (e) CMSCGC. (f) DGAE. (g) GCOT. (h) EKGCSC. (i) MS^{2}A-Net. (j) DSCRLE. (k) SDST. (l) SCMVC.

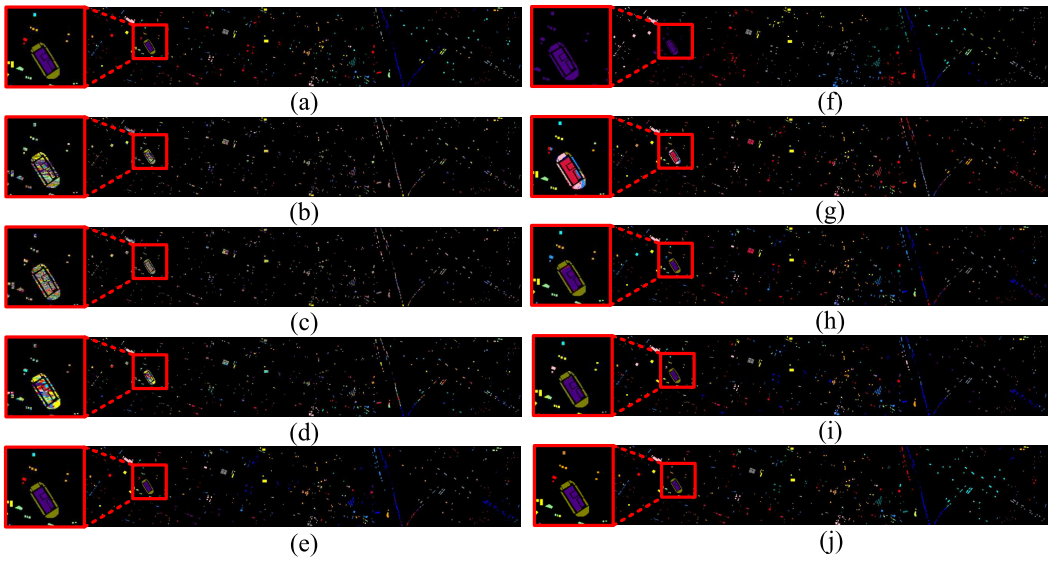


Fig. 7. Clustering maps of Houston 2013 image. (a) Ground truth. (b) MFLVC. (c) GCFAGG. (d) ACCMVC. (e) CMSCGC. (f) DGAE. (g) MS^{2}A-Net. (h) DSCRLE. (i) SDST. (j) SCMVC.

methods in identifying feature edges on the Indian-Pines dataset. Our method can better handle complex semantic information and preserve more details, unlike some methods (e.g., EGCSC, ACC: 54.63%) that lose a lot of detailed information. The clustering maps clearly demonstrate that our method has significant advantages in edge sharpness and continuity on the four HSI datasets. The clustering maps demonstrate that the proposed SCMVC can significantly enhance the discrimination separation for similar ground objects.

3) t-SNE Visualization: In this experiment, the t-distributed stochastic neighbor embedding (t-SNE) [57] is adopted to visualize the clustering feature distributions of Salinas, Botswana, Houston 2013, and Indian-Pines. The 2-D visualization of SCMVC on four datasets at epochs 0 and 200 is shown in Fig. 9. From the results, we can conclude that our SCMVC can well distinguish different types of samples. Specifically, the small interclass dissimilarity and the large intraclass variations existed in the original four datasets (first row). SCMVC can learn the most discriminative feature

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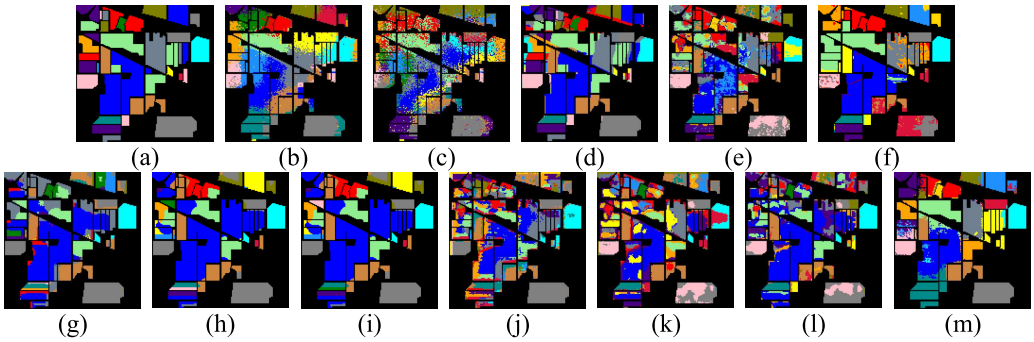


Fig. 8. Clustering maps of the Indian-Pines image. (a) Ground truth. (b) MFLVC. (c) GCFAGG. (d) ACCMVC. (e) CMSCGC. (f) DGAE. (g) GCOT. (h) EGCSC. (i) EKGCSC. (j) MS^{2}A-Net. (k) DSCRLE. (l) SDST. (m) SCMVC.

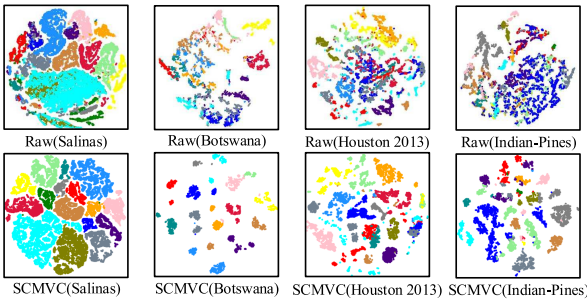


Fig. 9. 2-D visualization comparison between the feature distributions of SCMVC and the raw data.

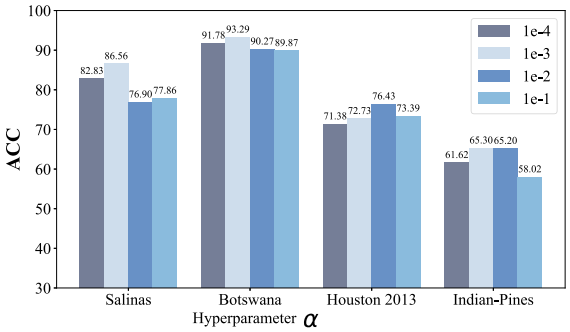


Fig. 10. ACC of our method with different α on the four HSI dataset.

representations, which are both intraclass compact and interclass dispersed, and contribute to higher clustering accuracy.

D. Parameter Analysis

SCMVC investigates several crucial parameters that have notable impacts on the clustering performance: parameters α, the learning rate, and the temperature τ. In this study, we employ cross-validation techniques to meticulously assess the sensitivity of each parameter. For each parameter, we provide corresponding variation curves of ACC on the Houston 2013, Botswana, Salinas, and Indian-Pines datasets.

1) Parameters α: In the proposed method, the overall loss function primarily has a key balancing parameter α. Fig. 10 presents the ACC of SCMVC on four different HSI datasets: Salinas, Botswana, Houston 2013, and Indian-Pines.

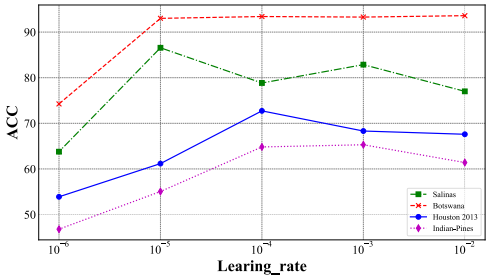


Fig. 11. ACC of our method with different learning_rate on the four HSI datasets.

The accuracies are shown for four different values of α: 1×10^{-4}, 1×10^{-3}, 1×10^{-2}, and 1×10^{-1}. For the Houston 2013 dataset, accuracy increases slightly as α ranges from 1×10^{-4} to 1×10^{-2}, but decreases slightly when α is 1×10^{-1}. For the other three hyperspectral datasets, the clustering accuracy is highest when $\alpha = 1\times10^{-3}$. As α increases, the proportion of semantic similarity consistency loss in the total loss gradually increases, causing SCMVC to lose its ability to distinguish between positive and negative sample pairs.

2) Learning Rate: Fig. 11 shows the performance of SCMVC on four different HSI datasets (Salinas, Botswana, Houston 2013, and Indian-Pines) with various learning rates (10^{-6} to 10^{-2}). Different datasets exhibit significant variations in performance at different learning rates. Houston 2013 performs best at a learning rate of 10^{-4}, but then drops slightly. Botswana reaches the highest accuracy at 10^{-5} and continues to improve at higher rates, remaining stable. Salinas performs best at a learning rate of 10^{-5}, while Indian-Pines performs best at 10^{-4}. Overall, learning rates of 10^{-5} and 10^{-4} yield better performance on most datasets, indicating that a moderate learning rate helps improve model accuracy.

3) Temperature τ: In semantic contrastive loss $\mathcal{L}_{SC}$, an appropriate temperature can help the model learn from hard negatives. Therefore, we analyze the impact of different temperatures on clustering in HSI data in Table VI. Our analysis shows that when $\tau = 0.5$, the clustering accuracy on three HSI datasets reaches its maximum value. While it achieves the best result with $\tau = 1$ on the Houston 2013 dataset, which has smaller spectral differences between classes, requiring a higher temperature τ to enhance discrimination of hard

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TABLE VI
CLUSTERING ACC(%) FOR MODELS TRAINED WITH DIFFERENT TEMPERATURE τ FOR CONTRASTIVE LOSS

Dataset	ACC/τ	ACC/τ	ACC/τ	ACC/τ
Houston 2013	60.97/(0.05)	64.69/(0.1)	73.43/(0.5)	76.43/(1)
Botswana	81.49/(0.05)	88.18/(0.1)	93.29/(0.5)	89.75/(1)
Salinas	66.86/(0.05)	75.43/(0.1)	86.56/(0.5)	74.81/(1)
Indian-Pines	45.70/(0.05)	52.96/(0.1)	65.30/(0.5)	59.77/(1)

TABLE VII
ABLATION STUDY RESULTS ON THE INDIAN-PINES DATASETS (%)

SCMVC	1	2	3	4	5
Spatial model	✓	×	✓	✓	✓
Spectral model	×	✓	✓	✓	✓
$\mathcal{L}_{SC}$	✓	✓	✓	×	✓
$\mathcal{L}_{SSC}$	✓	✓	×	✓	✓
Purity	68.26	41.14	41.62	75.86	75.52
Kappa	48.87	24.68	26.95	57.90	62.06
NMI	61.20	28.55	39.13	66.22	66.64
OA	54.52	31.87	41.36	61.16	65.30

negative examples. At an appropriate temperature, SCMVC increases the distance between negative samples and reduces the distance between positive samples.

E. Ablation Study

In this experiment, four key components, including spatial model, spectral model, semantic contrastive loss, and semantic similarity consistency loss, are investigated on the Indian-Pines dataset. The results are listed in Table VII. These methods are simplified from SCMVC-1 to SCMVC-5. The results reveal the necessity of each key component in SCMVC. Specifically, the clustering accuracy of SCMVC-5 is higher than that of SCMVC-1 and SCMVC-2, indicating that the spectral-enhanced spatial multiview feature is effective. Compared with SCMVC-4, the introduction of semantic contrastive loss enhances the consensus representation in SCMVC-5, producing better results for SCMVC, which confirms the important role of semantic contrastive loss. Similarly, the accuracy of SCMVC-5 is significantly better than that of SCMVC-3, demonstrating that the semantic similarity consistency loss enables SCMVC to produce a feature representation that is more favorable to clustering. Thus, the proposed spectral-enhanced spatial multiview feature, semantic contrastive loss, and semantic similarity consistency loss help to improve the clustering accuracy of SCMVC.

V. CONCLUSION

In this article, we proposed SCMVC, a SCMVC framework for HSIs. The method introduces an enhanced-view mechanism to reinforce similarity among related samples while increasing dissimilarity between distinct samples, effectively mitigating false negative pairs in multiview contrastive learning. We also designed a semantic constraint-based joint loss to capture high-level domain-invariant features and align semantically similar samples in feature space. By combining spectral-enhanced spatial multiview features with anchor-guided clustering, SCMVC achieves stable and discriminative cluster assignments. Extensive experiments on four benchmark HSI datasets demonstrate the effectiveness and general applicability of SCMVC, validating its superiority over existing state-of-the-art clustering methods.

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